

ABAYOMI D. NELSON

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PROFESSIONAL SUMMARY

Senior Data and AI professional with a Master of Data Science and a proven track record of architecting data models and pipelines that drive large-scale automation and scalable AI solutions. Expert in the full software development lifecycle (SDLC), specializing in translating complex business requirements into detailed technical designs and Jira-based work plans. Dedicated to leveraging Python, SQL, and AI-driven methodologies to scale data products and inform strategic decision-making in high-growth environments.

TECHNICAL SKILLS

Languages & Big Data: Python, PySpark, SQL, R, Hadoop, Delta Lake

Cloud & Architecture: Medallion Architecture, AWS Cloud Practitioner, Database Management, Docker

Data Processing & AI: Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG), Data Mining, ELT/ETL, Automation, Scripting

Tools & Methodologies: Power BI, Streamlit, Git Version Control, Jira, Agile Methodology, Project Management, Stakeholder Management, Business Intelligence, Data Visualization

AI & DATA ENGINEERING PROJECTS

Local Intel RAG Pipeline

- Architected a 100% private, local Retrieval-Augmented Generation (RAG) engine utilizing LangChain, ChromaDB, and Ollama (Llama 3).
- Implemented Maximal Marginal Relevance (MMR) retrieval logic to eliminate context bias and securely parse multi-page technical documents.

Global Aviation Data Pipeline (Medallion Architecture)

- Engineered a real-time ETL pipeline using PySpark and Delta Lake to ingest OAuth2-authenticated API telemetry processing thousands of flight vectors.
- Architected a three-tier Medallion structure (Bronze, Silver, Gold) with automated schema evolution and data type enforcement.

AutoMarket AI Pipeline

- Built an end-to-end predictive machine learning pipeline using Python and Scikit-Learn to automate automotive valuation models.
- Containerized the solution using Docker, ensuring highly scalable and reproducible environment deployments.

PROFESSIONAL EXPERIENCE

Epsilon

Chicago, IL | Feb 2025 – Present

Data Solutions Engineer

- Engineer and optimize petabyte-scale data pipelines using PySpark, Python, and the Hadoop framework, resulting in a 15% improvement in processing performance.
- Automate and manage end-to-end ETL/ELT solutions, reducing manual data tasks by 50% and increasing workflow efficiency.
- Manage the full software development lifecycle (SDLC) for data solutions, from converting business requirements into technical designs to estimation (JIRA), version control (Git), unit testing, and root cause analysis.

- Collaborate with stakeholders, architects, and QA teams to define and refine the big data platform, ensuring solutions align with strategic business objectives.

Chamberlain Group

Chicago, IL | Jan 2023 – Feb 2025

Data Product Analyst

- Served as the data product subject matter expert, collaborating with cross-functional teams to ensure timely and accurate data delivery by successfully launching 2 new data products within a quarter.
- Effectively translated business problems into technical solutions, which led to the widespread adoption of data-driven insights that resulted in significant business impact.
- Designed flexible and comprehensive dashboards that enabled the organization to answer the majority of their questions in minutes/hours instead of days/weeks by implementing the medallion architecture into the company's database.
- Increased data-driven decision-making by leveraging SQL and Power BI to extract actionable insights from customer behavior data. This resulted in a new marketing campaign, which led to a 10% increase in sales of the company's products.
- Conducted independent research on all aspects of a product's experiences and data, and provided findings and recommendations on which data are critical to fully understand the customer experience.
- Successfully onboarded new third-party data sources to improve business performance, which led to the acquisition of over 30 new corporate customers.
- Enhanced employee efficiency by 30% through the development and implementation of comprehensive data documentation. Reduced time spent by employees on data-related tasks by 5 hours per week.
- Established and implemented best practices for building analytical insights, ensuring data product quality and adoption. Reduced data errors by 35% and improved data integrity by 20%.
- Provided mentorship and guidance to team members and actively led team projects. Successfully mentored over 10+ team members in data manipulation and visualization.

Cadence Insurance

Baton Rouge, LA | July 2022 – Dec 2022

Data Analyst Intern

- Employed advanced data mining techniques, such as natural language processing and machine learning, to extract critical insights from over 500 TB of legacy data for 100+ clients, ensuring data integrity and accuracy.
- Developed and deployed a Django-based data management application that streamlined stakeholder access to data by 30%.
- Successfully led a data migration project that reduced data retrieval time by 50%, improving operational efficiency and enabling faster decision-making.
- Collaborated effectively with cross-functional teams to deliver data-driven solutions that increased client acquisition by 10% and improved client retention by 15%.

Louisiana State University - Dept. of Physics & Astronomy

Baton Rouge, LA | Apr 2022 – July 2022

Data Analyst Intern

- Utilized numerous Python Libraries to help users clean and analyze data, resulting in a 20% reduction in data processing time.
- Completed programming tasks by designing automation scripts, working with a total of 30 different scripts, leading to an 80% increase in user confidence in company data.
- Maintained user data by importing and exporting research data, resulting in a 15% improvement in data accuracy.

Louisiana Department of Treasury

Baton Rouge, LA | Aug 2021 – July 2022

Data Analyst Intern

- Processed extensive datasets utilizing SQL and Python, enhancing data integrity by 20% and facilitating precise analysis.
- Performed exploratory data analysis (EDA) on demographic data of Louisiana residents, resulting in valuable insights that shaped marketing strategies and boosted customer retention by 15%.

- Designed and automated reporting procedures using Tableau, resulting in a 50% reduction in weekly reporting time and improved team efficiency.

EDUCATION & TRAINING

Master of Data Science | Eastern University | 2025

Bachelor of Science in Information Systems & Decision Sciences | Louisiana State University | 2022

Certifications: SAFe Scrum Master (2024), Amazon Web Services Cloud Practitioner (2022)